

Ali Awil

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OBJECTIVE

Computer Science graduate (Data Science option) from the University of Ottawa with hands-on experience in machine learning, AI model evaluation, and data-driven software development. Skilled in building intelligent, practical systems that bridge research and real-world impact.

EDUCATION

Bachelor of Computer Science (Data Science Option)

University of Ottawa, Ottawa, ON (Expected Dec 2025)

GPA 9.34 / 10 (3.7 / 4) | Dean's List (All Semesters)

Relevant Courses: Artificial Intelligence, Machine Learning, Fundamentals of Data Science, Information Retrieval & Internet, Data Structures and Algorithms, Design and Analysis of Algorithms

SKILLS

- **Languages:** Python, Java, C++, Prolog, JavaScript, TypeScript, SQL, R
- **Frameworks / Tools:** React, LangChain, Flask, FastAPI, PyTorch, Scikit-learn, Hugging Face, Power BI, Docker, Azure, Git
- **Concepts:** Machine Learning, Data Analysis, Model Evaluation, ETL, Vector Databases, Cloud Deployment, Data Visualization

EXPERIENCE

Data Science Developer Intern | Department of National Defence, Ottawa, Canada

May 2024 - May 2025

- Worked on two major AI projects, LLaVA and Space News, documenting the full process from data preparation to testing in internal research papers.
- Tested a fine-tuned LLaVA (WizardLM-70B + CLIP) model and analyzed outputs to evaluate multimodal reasoning quality.
- Compared Longwizard-70B and Mistral Nemo Instruct models on real FAM policy Q&A, measured accuracy, latency, and improvement potential.
- Evaluated the Space News pipeline, checked data quality, and verified extracted details such as locations, dates, rockets, and satellites.

Software Engineer Intern | Zafin, Toronto, Canada

May 2023 - August 2023

- Built an Enterprise Knowledge Assistant using OpenAI's API that achieved 90% accuracy on internal test data.
- Scraped and processed internal documentation and FAQ data to create a structured knowledge base for model training and retrieval.
- Stored embeddings in a PostgreSQL vector database and implemented retrieval using LangChain and OpenAI function calling.
- Evaluated Falcon-7B (Hugging Face) alongside OpenAI GPT models to assess open-source vs. proprietary performance and cost trade-offs.
- Developed a Flask microservice for model queries and API calls, deployed on Microsoft Azure.
- Co-authored "[Closing the Gap Between Available Information and Accessible Knowledge](#)", a company research article published on Zafin's Engineering Blog (Medium, Aug 2023).

PROJECTS

Customer Behaviour Analysis in Retail (Python, SQL, Power BI)

- Designed and implemented a data mart and ETL pipeline to analyze customer shopping trends.
- Built OLAP queries and Power BI dashboards to visualize purchasing patterns and business insights.
- Applied data mining techniques (clustering and classification) to identify customer segments and predict buying behaviour.

Information Retrieval System (Python)

- Implemented a multithreaded TF-IDF-based document retrieval system with preprocessing, inverted indexing, and cosine similarity ranking.
- Evaluated performance on 140K+ token corpus using TREC Eval, achieving MAP = 0.27 (titles-only) vs 0.19 (titles + descriptions).

Bookstore Database Application (Java, SQL)

- Built a database system to track inventory and sales with optimized CRUD operations and query performance.

Student Course Booking App (Java, Firebase)

- Developed a web application for course registration and schedule management with real-time data storage.

Open to relocation across Canada and the U.S.